PHYS 225

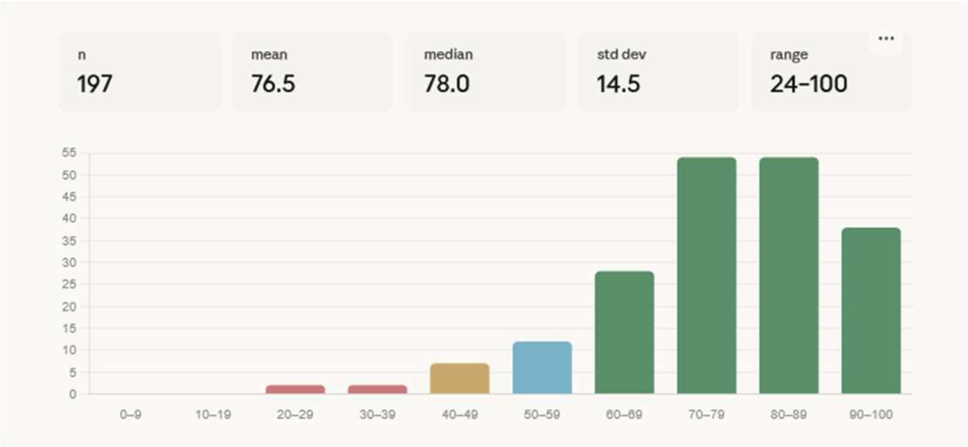
Week 10, Lecture 8: Relativistic energy, momentum and force

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Reminders

- Welcome back! Normal schedule resumes this week.

- **Midterm update**: Raw scores are posted in the Gradebook. Good job everyone!



- Office Hours Wednesday, Thursday usual time.
- Homework 8 assigned this Thurs. 10/23 due next Thurs. 10/30.
- Don’t forget to turn in your **1-minute papers** by 6 pm today!!!

Week 10, Lecture 8

Relativistic energy, momentum, and force

Objectives:

- Define the relativistic versions of momentum, force, and energy
- Understand Einstein's famous equation $E = mc^2$

One Minute Paper: define force and energy in classical mechanics in terms of momentum, mass, and/or time derivatives

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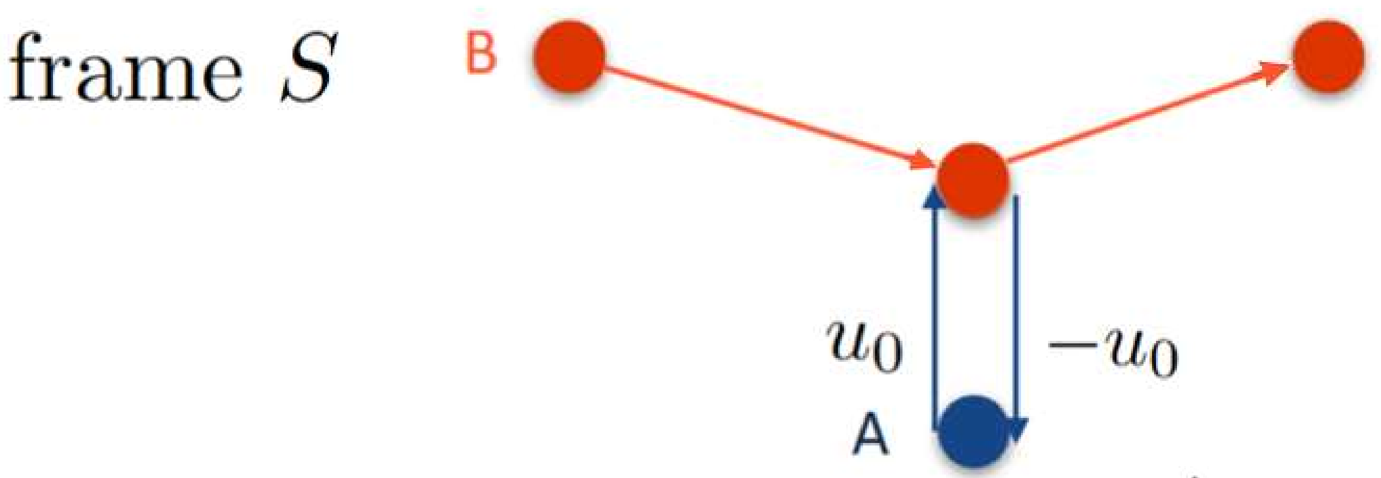
Relativistic momentum

Momentum is useful because it is **conserved** in the absence of external forces. We want to define momentum in special relativity so that it remains conserved at relativistic speeds.

Consider two identical balls of mass m, colliding as follows:



$$\Delta p_{y,A} = -2mu_0$$

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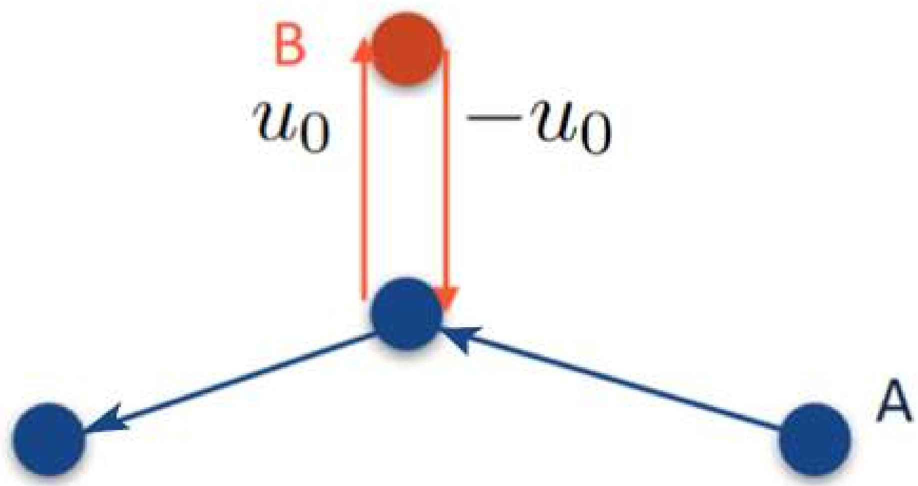
Relativistic momentum

Momentum is useful because it is **conserved** in the absence of external forces. We want to define momentum in special relativity so that it remains conserved at relativistic speeds.

Let's look in a frame traveling with B's x-velocity:

frame S'



This is just the same thing A sees, but upside down and backwards.
Therefore it is consistent for B to have speed u_0 in this frame.

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Relativistic momentum

Momentum is useful because it is **conserved** in the absence of external forces. We want to define momentum in special relativity so that it remains conserved at relativistic speeds.

What is the change in B's momentum in frame S?

$\text{frame } S$

Transverse velocity addition (Discussion 5):

$$u_{B,S} = \frac{u_0}{\gamma}$$

$$\Delta p_{y,B} = 2mu_0/\gamma$$

$$\Delta p_y = \Delta p_{y,A} + \Delta p_{y,B} \neq 0!$$

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Relativistic momentum

How can we fix this? **Define** the relativistic momentum as

$$\boxed{\mathbf{p} = \gamma m \mathbf{v}}$$

$$\frac{1}{\sqrt{1 - v^2/c^2}}$$

v is particle's speed

mass of the object

velocity of an object in a given frame

[blackboard]

Reduces to Newtonian value at small velocities because $\gamma \xrightarrow{v \to 0} 1$

A's velocity in S is $+u_0\hat{y}$ before collision and $-u_0\hat{y}$ after collision, so its relativistic momentum is

$$\vec{p}_{A,1} = \gamma_u m u_0 \hat{y}$$
$$\vec{p}_{A,2} = -\gamma_u m u_0 \hat{y} \quad \Rightarrow \Delta p_{A,y} = -2\gamma_u m u_0 \quad \left(\gamma_u = \frac{1}{\sqrt{1-u_0^2/c^2}}\right)$$

B's velocity in S' is $-u_0\hat{y}$ before and $+u_0\hat{y}$ after.

If S' has velocity $v\hat{x}$, then $v_{B,x} = v$ and $v_{B,y} = \mp \frac{u_0}{\gamma_v}$

w/ $\gamma_v = \frac{1}{\sqrt{1-v^2/c^2}}$, so B's speed in S is [Discussion 5]

$$v_B = \sqrt{v_{B,x}^2 + v_{B,y}^2} = \sqrt{v^2 + u_0^2\left(1-\frac{v^2}{c^2}\right)}$$

B's γ factor is

$$\gamma_B = \frac{1}{\sqrt{1-v_B^2/c^2}} = \frac{1}{\sqrt{1-\frac{v^2}{c^2}-\frac{u_0^2}{c^2}+\frac{u_0^2v^2}{c^4}}} = \frac{1}{\sqrt{\left(1-\frac{u_0^2}{c^2}\right)\left(1-\frac{v^2}{c^2}\right)}} = \gamma_u\gamma_v$$

$$\Rightarrow \vec{p}_{B,1} = \gamma_u\gamma_v m\left(v\hat{x}-\frac{u_0}{\gamma_v}\hat{y}\right)$$

$$\vec{p}_{B,2} = \gamma_u\gamma_v m\left(v\hat{x}+\frac{u_0}{\gamma_v}\hat{y}\right)$$

$$\Rightarrow \Delta p_{B,y} = \frac{2\gamma_u\gamma_v m u_0}{\gamma_v}$$

$$= 2\gamma_u m u_0$$

$$= -\Delta p_{A,y} \checkmark$$

Relativistic momentum is conserved!

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Relativistic force

Now that we know how to define momentum, let's try force:

$$\boxed{\mathbf{F} = \frac{d\mathbf{p}}{dt}}$$

relativistic
momentum

Note: all quantities (including time!) measured in same frame

Reduces to Newtonian value at small velocities because **p** does, and all frames in Newtonian mechanics have the same time

Week 10, Lecture 8

Relativistic force

Now that we know how to define momentum, let's try force:

$$\boxed{\mathbf{F} = \frac{d\mathbf{p}}{dt}}$$

relativistic momentum

1D.

$$\frac{d}{dt}(\gamma m v) = m\left(\dot{\gamma} v + \gamma \dot{v}\right)$$

$$\dot{\gamma} = \frac{d}{dt}\left(\frac{1}{\sqrt{1-v^2/c^2}}\right) = -\frac{1}{2}\frac{-2v/c^2}{(1-v^2/c^2)^{3/2}}\dot{v} = \gamma^3 va/c^2$$

$$= m\left(\frac{\gamma^3 v^2 a}{c^2} + \gamma a\right) = \gamma^3 ma\left(\frac{v^2}{c^2} + \frac{1}{\gamma^2}\right) = \gamma^3 ma$$

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Relativistic energy

Finally, let's relate energy to force through the work-energy theorem:

$$\Delta E = \int \mathbf{F} \cdot d\mathbf{l}$$

1D.

will get to line integrals in a couple weeks...

$$\int_0^{x_f} F(x)\,dx = \int_0^{x_f} (\gamma^3 ma)\,dx$$

$$a = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}$$

$$= \int_0^{x_f} m\gamma^3 v \frac{dv}{dx} dx$$

$$= \int_0^{v_f} m\gamma^3 v\,dv$$

Change variables one more time...

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Relativistic energy

$$\int_0^{x_f} F\,dx = \int_0^{v_f} m\gamma^3 v\,dv \qquad \frac{d\gamma}{dv} = \frac{d}{dv}\left(\frac{1}{\sqrt{1-v^2/c^2}}\right) = -\frac{1}{2}\left(\frac{-2v/c^2}{(1-v^2/c^2)^{3/2}}\right) = \frac{v\gamma^3}{c^2}$$

$$= mc^2 \int_{\gamma_0}^{\gamma_f} d\gamma \qquad \gamma_0 = 1$$

$$= mc^2(\gamma_f - 1)$$

In other words, $\boxed{\text{K.E.} = (\gamma - 1)mc^2}$

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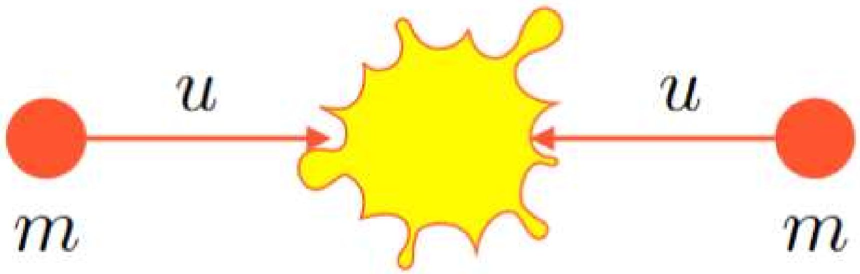
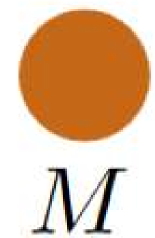
Relativistic energy

Consider collision



two particles of mass m collide...



creating one particle of mass M

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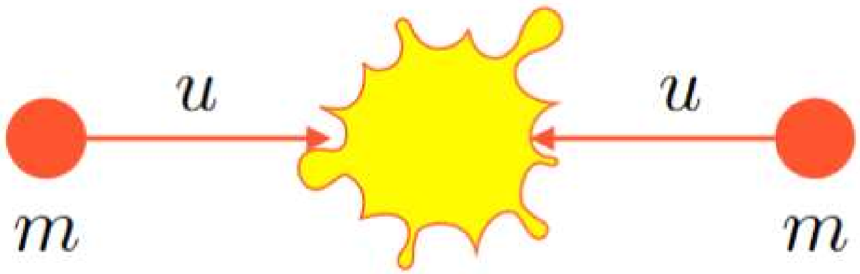
Relativistic energy

Consider collision



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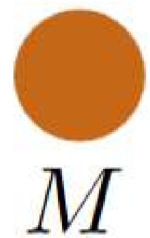

creating one particle of mass M

Consider frame moving to the left with velocity u:

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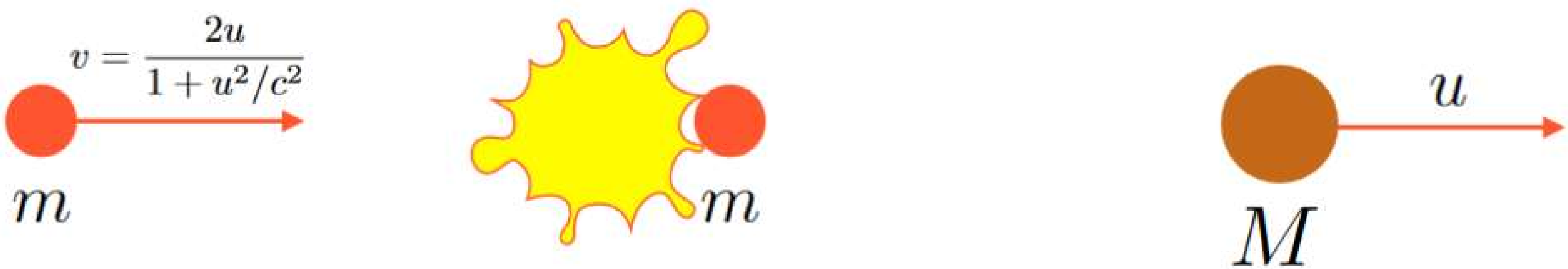
Relativistic energy



two particles of mass m collide... creating one particle of mass M

$$\gamma_v = \frac{1 + u^2/c^2}{1 - u^2/c^2}$$ (Morin 3.1.2, or check for yourself!)

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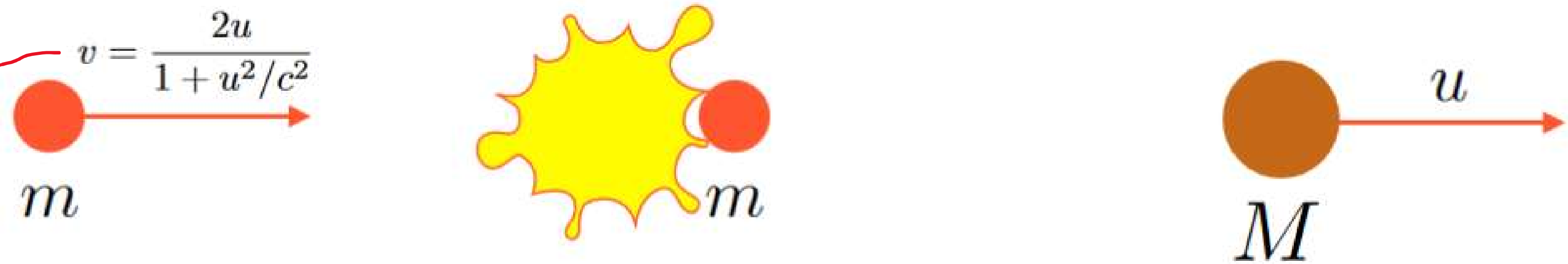
Relativistic energy



two particles of mass m collide… creating one particle of mass M

$$\gamma_v = \frac{1 + u^2/c^2}{1 - u^2/c^2}$$ (Morin 3.1.2, or check for yourself!)

Apply **relativistic** momentum conservation:

$$\gamma_v m v = \gamma_u M u \implies m\left(\frac{1 + u^2/c^2}{1 - u^2/c^2}\right)\left(\frac{2u}{1 + u^2/c^2}\right) = \frac{Mu}{\sqrt{1 - u^2/c^2}}$$

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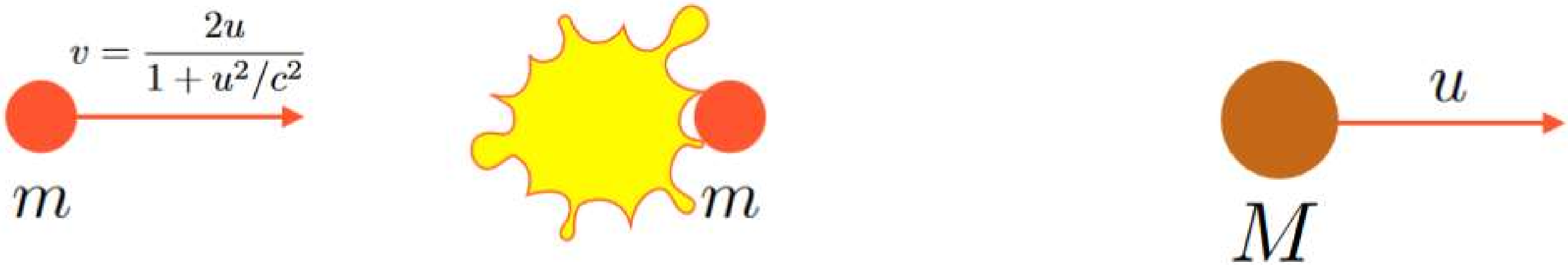
Relativistic energy



two particles of mass m collide... creating one particle of mass M

$$\gamma_v = \frac{1+u^2/c^2}{1-u^2/c^2}$$ (Morin 3.1.2, or check for yourself!)

$$M = 2\gamma_u m$$

mass is not conserved!

Apply **relativistic** momentum conservation:

$$\gamma_v m v = \gamma_u M u \implies m\left(\frac{1+u^2/c^2}{1-u^2/c^2}\right)\left(\frac{2u}{1+u^2/c^2}\right) = \frac{Mu}{\sqrt{1-u^2/c^2}}$$

$$2mu\gamma_u^2 = \gamma_u M u$$

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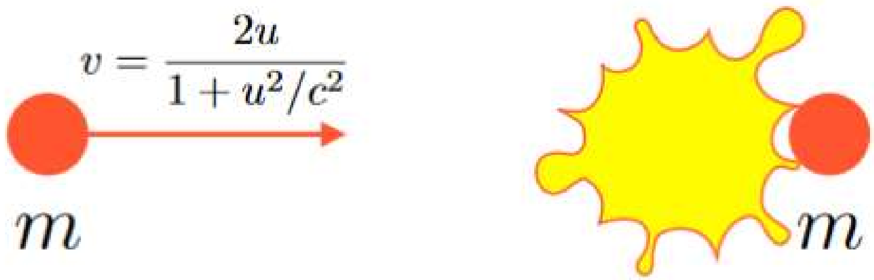
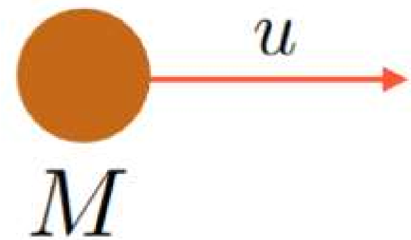
Relativistic energy



two particles of mass m collide...



creating one particle of mass M

$$\gamma_v = \frac{1+u^2/c^2}{1-u^2/c^2}$$ (Morin 3.1.2, or check for yourself!)

$$M = 2\gamma_u m$$

mass is not conserved!

Relativistic energy **is** conserved (Morin 3.1.2):

$$\boxed{E = \gamma m c^2}$$

Week 10, Lecture 8

Einstein's most famous equation

When we took the low-velocity (Newtonian) limit of relativistic momentum and force, we got

$$\mathbf{p} = \gamma m \mathbf{v} \quad \Rightarrow \quad \mathbf{p}_0 = m\mathbf{v}$$

$$F = \gamma^3 m a \quad \Rightarrow \quad F_0 = ma$$

When we try this for energy, we get

$$E = \gamma mc^2 \quad \Rightarrow \quad \boxed{E_0 = mc^2}$$

A particle at rest still has an energy E_0, called its rest energy, that can be exchanged during reactions (week 13)

Week 10, Lecture 8

Relativistic energy, momentum, and force

Summary

- Relativistic momentum: $\mathbf{p} = \gamma m \mathbf{v}$
- Relativistic energy: $E = \gamma m c^2$
- Newton's Second Law still holds as $\mathbf{F} = d\mathbf{p}/dt$
- Rest energy: $E_0 = mc^2$

One Minute Paper: which of your equations from the start-of-class One Minute Paper continues to hold in special relativity?